

If Statements

In this part of the lab, we're going to combine what we learned in the last 2 parts, the way we're going to use it is the normal way you'd use these topics.

But first let's go over if statements

To put it simply, an if statement just means checking a CONDITION. Something as simple as "Do you have enough money to buy this product?" If not, tell them you can't. If they do **give the product**.

- Before we go into if statements you need to understand how to make comparison.
 - You should remember $<$, $>$, $=$, $\leq$, $\geq$ this is just for comparing two values (usually numbers, *little note for comparing values with $=$, we use double $==$.*)
 - In coding languages the `!"` is used to flip an instruction, *i.e. make it NOT that*. So `!=` is for checking if two numbers are NOT equal to each other.
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If statement, check the condition, is it true that we have enough to pay?

```
payment = 10
cost = 15
if payment <= cost: # if value (some comparison to) other value
    print("Here's your item.")
```

What if we DON'T meet it? If we only have TWO scenarios YES & NO. Then we can just have something called an `"else"`. This is because the payment can EITHER be "greater or equal to payment cost" or NOT.

So anything that doesn't meet our first comparison means it HAS to fall into the 2nd one.

- Meaning, unlike our if statement, we do NOT need a condition

```
payment = 10
cost = 15
if payment <= cost: # if value (some comparison to) other value
    print("Here's your item.")
else:
    print("Can't buy.")
```

What if I don't just want a YES or NO? Like what if the payment is like 0.05 short? I'd want to allow that but person to buy, maybe add to a variable called "loss" that adds up the loss.

- We'd have to add that IN BETWEEN our else, since **else is supposed to catch anything that isn't considered**. We add a new if statement between as to say "If my first statement, and 2nd statement are false, then do that."

This is called an **elif (else if)**, exact same as if but it just comes after the an if statement, *in fact we can have AS MANY elif's as we want.*

```
payment = 10
cost = 15
loss = 0
if payment <= cost: # if value (some comparison to) other value
    print("Here's your item.")
elif cost - payment < 0.05:
    loss += cost - payment
else:
    print("Can't buy.")
```

The line "loss += cost - payment" is probably new to you. += is a special tool, it means take whatever value that is already saved in here, and add

- Let's say I payed with 3 cents less, **loss** now equals 0.03. If I do it again and I payed with 2 cents less, **loss** is now 0.03 + 0.02. So 0.05.
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Now, let's add our LISTS and INPUTS.

- Quick tip, **LISTS and INPUTS** and generally any variable should be made BEFORE any if statements.

Our list should have the **COST**, and the **PRODUCT NAME**;

```
products = {"apple": 0.5, "banana": 0.3, "orange": 0.4}
```

Then let's add our inputs; then our LOSS value

```
payment = float(input("Enter payment amount: "))
product_choice = input("Enter product choice (apple, banana, orange): ")
loss = 0.0
```

The program should start with this.

```
products = {"apple": 0.5, "banana": 0.3, "orange": 0.4}
```

```
payment = float(input("Enter payment amount: "))
```

```
product_choice = input("Enter product choice (apple, banana, orange): ")
```

```
loss = 0.0
```

Then we go into our if statement, we are going to **USE our list (Dictionary)** to do our comparison.

- We will compare `payment`, to the value that is attached to the `product_choice` made by the user.
- Remember we **ACCESS** the value in a dictionary by using it's "key" which is it's name.
 - `products[product_choice]`, Assume the user put in "apple" for `product_choice`, this will be 0.5 then, same as `products["apple"]`
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Continue; and end off our program like this; *All we've changed from before is REPLACE "payment" with an input; And grab the cost from a dictionary. Otherwise this is the SAME thing.*

```
if payment <= products[product_choice]:
    print("Here's your item.", product_choice)
elif products[product_choice] - payment < 0.05:
    loss += products[product_choice] - payment
    print("Here's your item.", product_choice)
else:
    print("Can't buy.")
```

Combine both of those snippets of code above to make our program. **TO RECAP:**

- In general when making a program. You want to have your lists, inputs and variables made **before your IF statements**.
- If statements can have unlimited "else if" conditions, the else catches anything not covered. (Like, maybe checking if a grade is A,B,C,D by comparing to the values for each grade.)
- **If statements** are used to go through different paths, based on inputs.

Next part will cover two concepts, FOR loops which is used to go through every item in a list. **An objects**, which are just dictionaries again `{}`.